

Dictionary Sinicum: Building a 428K-Headword Chinese Dictionary in 12 Languages with a Single LLM for Under \$100

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Abstract

We present *Dictionary Sinicum*, an open multilingual Chinese dictionary comprising 428,073 headwords with definitions in 12 languages, constructed by merging seven community-maintained dictionaries with glosses generated by a single LLM (MiniMax M2.5) at a total inference cost of ~\$100. A three-phase pipeline—batch translation, retry, and context-aware backfill—achieves 100% coverage across all 5.1M headword–language slots. We describe prompt engineering techniques that prevent script contamination in CJK-target languages, including Japanese kanji echo and Korean hanja mixing. Automated comparison against four community reference dictionaries over 453 headword–language pairs shows 87.3% sense coverage with a 12.5% false sense rate. The dictionary, augmented with 184K Cantonese and Hokkien dialect forms, is released under CC BY-SA 4.0.

1 Introduction

Bilingual and multilingual dictionaries remain essential infrastructure for language learning, translation, and NLP. For Chinese, the open-source community has produced high-quality dictionaries for a handful of language pairs—CC-CEDICT for English (~124K entries), HanDeDict for German (~159K entries), CFDict for French (~56K entries), and CC-CIDICT for Indonesian (~124K entries)—but no comparable resources exist for Spanish, Korean, Russian, Vietnamese, Tagalog, Persian, or Swedish. Creating such dictionaries manually requires years of expert effort per language pair.

Recent advances in large language models suggest an alternative: using a single multilingual model to generate dictionary-quality definitions at scale. However, prior work on LLM-assisted lexicography has been limited to small-scale experi-

ments. The closest precedent is a Pre-Qin philosophy lexicon project using Qwen3-14B to generate bilingual dictionary entries for approximately 1,000 classical Chinese philosophical terms in a single language pair (Authors redacted for review, 2026). Lu et al. (2024) use existing bilingual dictionaries as chain-of-thought context to improve LLM translation of rare words—a complementary approach where dictionaries help LLMs translate, whereas we use LLMs to *build* dictionaries. No published work has attempted dictionary generation at the scale of hundreds of thousands of headwords across more than two target languages.

We address this gap with *Dictionary Sinicum*, a system that:

1. **Merges** seven community-maintained Chinese dictionaries covering 6 language pairs into a unified headword table of 428,073 entries.
2. **Translates** all headwords into 12 languages using a single LLM with carefully engineered prompts that produce dictionary-style output and prevent script contamination.
3. **Backfills** gaps using a context-aware prompt that sends existing translations as read-only context, ensuring consistency across languages.
4. **Augments** the dictionary with 184,000 dialect forms covering Cantonese (Jyutping) and Hokkien (POJ) pronunciations and lexical equivalents.

The total inference cost was approximately \$100, and the full pipeline completed in under one week. The project’s name honors the *Dictionario Hispanico Sinicum* (1626–1642), a 27,000-entry Spanish-

Chinese-Hokkien dictionary compiled by Dominican missionaries in Manila—a reminder that multilingual Chinese lexicography has a four-century history, and that the Hokkien maritime vocabulary in our dialect data traces the same trade routes documented in that manuscript.

2 Related Work

2.1 Multilingual Lexical Resources

Large-scale multilingual lexical resources have been constructed through both manual and automated methods. PanLex (Kamholz et al., 2014) aggregates bilingual dictionaries into a graph covering ~5,700 languages with over 1.2 billion pairwise translations, but its Chinese coverage is limited to headword-level mappings without definition glosses. BabelNet (Navigli and Ponzetto, 2012) combines WordNet, Wikipedia, and other resources into a multilingual encyclopedic dictionary, but is not open-access for bulk redistribution and focuses on synset alignment rather than dictionary-style glosses. Our work differs from both in producing complete, human-readable definition glosses for a single source language across 12 targets, released under CC BY-SA 4.0.

2.2 LLM-Assisted Lexicography

The use of LLMs for dictionary construction is an emerging area with limited published work. The closest precedent is a Pre-Qin philosophy lexicon (Authors redacted for review, 2026) that uses Qwen3-14B to generate ~1,000 bilingual entries for classical Chinese philosophical terms—two orders of magnitude smaller than our work. Chain-of-Dictionary prompting (Lu et al., 2024) uses existing bilingual dictionaries as chain-of-thought context to improve LLM translation of rare words—a complementary approach where dictionaries help LLMs translate, whereas we use LLMs to *build* dictionaries.

2.3 Multilingual Translation Models

General-purpose multilingual translation models such as NLLB (Costa-jussà et al., 2022) cover 200+ languages but are designed for sentence-level translation, not dictionary-style glosses. Tower (Alves et al., 2024) focuses on high-resource language pairs. Neither has been evaluated for or applied to dictionary construction. These models do not natively produce multi-sense gloss lists, making direct comparison nontrivial (see Limitations).

Source	Lang	Entries	License
CC-CEDICT	en	123,127	CC BY-SA 4.0
HanDeDict	de	159,809	CC BY-SA 3.0
CFDICT	fr	56,280	CC BY-SA 3.0
CC-CIDICT	id	123,157	CC BY-SA 4.0
CHEDICC	es	4,400	CC BY-SA 4.0
Wiktextextract	multi	140,577	CC BY-SA 4.0
JMdict	ja	131,427	CC BY-SA 4.0

Table 1: Source dictionaries. All use compatible Creative Commons licenses.

2.4 Community Chinese Dictionaries

The open Chinese dictionary ecosystem is built on the CC-CEDICT model: community-maintained, CEDICT-format text files released under Creative Commons licenses. CC-CEDICT (ced, 2024), HanDeDict (han, 2024), CFDICT (cfd, 2024), and CC-CIDICT (cid, 2024) together cover four European languages and Indonesian. Wiktextextract (Ylonen, 2024) provides multilingual glosses extracted from Wiktionary, and JMdict (Breen, 2024) covers Japanese. No comparable community dictionaries exist for Spanish, Korean, Russian, Vietnamese, Tagalog, Persian, or Swedish.

3 Data Sources and Headword Unification

3.1 Source Dictionaries

Table 1 lists the seven source dictionaries. All sources use compatible Creative Commons licenses (BY-SA 3.0/4.0), allowing the combined work to be distributed under CC BY-SA 4.0.

3.2 Headword Unification

Source dictionaries use varying conventions for pinyin romanization, traditional/simplified character pairs, and entry granularity. We unified headwords on the triple (traditional, simplified, pinyin), yielding 428,073 unique headwords. Of these, 30,546 are single-character entries covering 19,471 distinct Chinese characters (many characters have multiple pinyin readings—e.g., 参 has 12 distinct entries).

3.3 Dialect Forms

We augmented the dictionary with Cantonese and Hokkien dialect data from three CC-licensed sources (Table 2). Hokkien romanization uses two systems: iTaigi provides Tâi-lô (the MOE standard), while TaiHua uses traditional POJ (Pe h-ōe-jī). A distinctive feature of Hokkien is its high lexical divergence from Mandarin: 68–78% of Hokkien

Source	Dialect	Forms	System
CC-Canto	Cantonese	125,640	Jyutping
iTaigi	Hokkien	18,762	Tâi-lô
TaiHua	Hokkien	40,051	POJ

Table 2: Dialect data sources. Total: 184,453 forms across 121,045 headwords with Cantonese and 53,684 with Hokkien coverage.

Model	Provider	s/ent	Issues
MiniMax M2.5	MiniMax	1.2	None
Kimi K2	Groq	1.8	None
Nemotron 253B	NIM	7.0	None
Sonnet 4.6	Anthropic	19.8	HW echo
DeepSeek V3.2	NIM	35.3	HW echo
MiniMax M2.1	NIM	167	Think ON
MiniMax M2.1	Fireworks	5.4	None

Table 3: Model evaluation summary (5-headword test suite). HW echo = headword echo in output. Think ON = thinking mode silently enabled.

forms use different characters entirely (e.g., 漂亮 *piàoliang* “beautiful” → 嬌 *suí*; 東西 *dōngxi* “thing” → 物件 *mi h-kiānn*).

4 Translation Pipeline

4.1 Model Selection

We evaluated 20+ LLMs across 10 inference providers on a 5-headword test suite covering emergency service numbers (culturally specific), anime titles (transliteration challenges), and medical terminology. Selection criteria were: (1) correct 12-language output in the prescribed format, (2) no script contamination, (3) geographic context preservation, (4) sustainable throughput at 428K headwords, and (5) cost under \$200.

MiniMax M2.5 was selected for production: zero issues on the test suite, 1.2s/entry throughput, no observed rate limits with 20 parallel workers, and an estimated cost of \$93 for the full run (Table 3).

A notable finding is that the same open-weight model can produce dramatically different output depending on the inference provider. MiniMax M2.1 served by NVIDIA NIM took 167s per entry with thinking mode silently enabled, while the same weights on Fireworks took 5.4s with thinking disabled. Benchmark scores (SWE-bench, MMLU) showed no predictive value for this structured multilingual task. These observations align with emerging evidence that inference acceleration introduces unpredictable behavioral changes (Kirsten et al., 2025).

Phase	Entries	Defs	Time	Cost
1. Main	374,602	4.41M	24h	\$93
2. Retry	53,471	616K	92m	\$3
3. Backfill	82,550	110K	105m	\$4
Total	428,073	5.14M	27h	\$100

Table 4: Three-phase execution summary.

4.2 Prompt Engineering

The production prompt generates all 12 languages simultaneously for each batch of 20 headwords. Key design decisions:

System prompt (176 tokens): Establishes the lexicographer persona, output format (xx: def1/def2), conciseness constraints (max 5 glosses), and language-specific rules addressing three classes of script contamination:

- Japanese kanji echo:** Without explicit instruction, LLMs frequently “define” a Chinese word in Japanese by repeating the same characters (e.g., 銀行 → 銀行). The prompt requires meaning explanations, not kanji echo.
- Korean hanja mixing:** Korean definitions often include Chinese characters (hanja) unrecognizable to most modern Korean readers. The prompt specifies Hangul only.
- General CJK leakage:** A blanket prohibition on Chinese characters in non-Chinese definitions.

User prompt (batch of 20): Each entry includes traditional/simplified characters, pinyin, POS tag, and all existing community definitions as context. This constitutes **pivot translation**: for the six languages without community dictionaries (Spanish, Korean, Russian, Vietnamese, Tagalog, Persian, Swedish), the model effectively translates from the English and other community-language glosses rather than directly from Chinese.

Token economics : ~5,500 tokens per batch (500 system + 2,000 input + 3,000 output). Full run: 21,403 batches.

4.3 Three-Phase Execution

Table 4 summarizes the three-phase execution. **Phase 1** processed all headwords in batches of 20 with 20 parallel workers. The API returned unparseable responses for 53,471 headwords (12.5%). **Phase 2** retried failed headwords, recovering 616K

additional definitions. **Phase 3** used a distinct context-aware backfill prompt that sends all existing translations as read-only context, asking the model to produce only the missing languages. This design ensures consistency: the model sees existing glosses and produces output that matches in style and specificity.

4.4 Parser Robustness

Parsing structured LLM output at scale required handling several failure modes: multi-language line collapse (the model outputs multiple languages on one line), numbered vs. unnumbered responses, and overwrite protection via three-layer guards (query filter, parser filter, database existence check).

5 Quality Assessment

5.1 CJK Leak Audit

We sampled 10,000 definitions per non-Japanese language (110,000 total) and measured CJK character contamination. The overall leak rate across non-Japanese languages was **2.49%** of sampled definitions (weighted). Korean showed the highest rate (0.79%) due to hanja mixing, followed by Tagalog (0.42%) and Vietnamese (0.38%).

This metric conflates three distinct phenomena: (1) **genuine errors**—Chinese text serving no function in the target language; (2) **conventional script mixing**—Korean hanja usage is standard in Korean technical dictionaries; Vietnamese Hán-Việt vocabulary is a fundamental lexical layer, not contamination; (3) **metalinguistic references**—cross-reference entries necessarily cite Chinese characters. A cleanup script targets category 1 errors; categories 2 and 3 require language-specific policies.

5.2 Comparison with Community Dictionaries

For the four languages with existing community dictionaries providing glosses (English, German, French, Indonesian), we compared LLM-generated definitions against human-curated references.¹ We sampled 200 headwords stratified by community source coverage as a frequency proxy and measured two metrics across 453 headword–language pairs:

1. **Sense coverage:** proportion of reference senses captured by the LLM gloss (automated lexical overlap with 50% word-overlap threshold)

¹JMdict was excluded because it stores kana readings rather than Japanese-language glosses.

Lang	Reference	n	Cov.	False
en	CC-CEDICT	139	85.8%	14.1%
de	HanDeDict	105	87.9%	17.9%
fr	CFDICT	70	81.2%	18.6%
id	CC-CIDICT	139	91.3%	3.7%
Overall		453	87.3%	12.5%

Table 5: LLM vs. community dictionary comparison. Cov. = sense coverage; False = false sense rate.

Band	n	Cov.	False
high (4+ sources)	190	84.7%	16.7%
mid (3 sources)	146	87.4%	11.8%
low (2 sources)	94	90.8%	5.7%
rare (1 source)	23	93.5%	10.9%

Table 6: Comparison by frequency band (community source coverage as proxy).

2. **False sense rate:** proportion of LLM glosses not matched by any reference sense

Important caveat. The translation prompt provides all existing community definitions as input context (§4). For these four languages, the model *sees* the reference glosses and is asked to produce output in the same language. This comparison therefore measures **context utilization**—how faithfully the model preserves reference senses—not independent translation quality. We find that 42% of English, 40% of German, 55% of French, and 60% of Indonesian LLM glosses are substring matches of the corresponding community definition.

The overall sense coverage of 87.3% (95% bootstrap CI: [84.7%, 89.8%]) with a false sense rate of 12.5% (95% CI: [10.1%, 15.0%]) shows that the model successfully preserves most reference senses while introducing few unsupported glosses (Tables 5–6). Manual inspection reveals that most “false senses” are legitimate synonyms absent from the reference.

The main failure mode is **sense compression**: the LLM produces fewer glosses than community dictionaries (median 2 vs. 3), preferring the most common translation equivalent and omitting specialized or archaic senses. This raises a linguistically interesting question: does the model compress senses uniformly, or does it systematically drop certain sense types (archaic, domain-specific, colloquial)? Preliminary inspection suggests the latter, but a controlled study with sense-type annotation is needed.

For the six languages *without* community dictionaries, the model relies on pivot translation and we

have no external reference for quality assessment.

5.3 Source Priority and Definition Merging

For headwords with definitions from multiple sources, we apply a priority hierarchy: community dictionaries > Wikitexttract > MiniMax. Human-curated definitions take precedence, with LLM translations filling gaps for languages without community dictionaries.

5.4 Coverage Summary

Every headword–language slot ($428,073 \times 12 = 5,136,876$) has an LLM-generated gloss. For the 739,305 slots also covered by a community dictionary, both definitions are stored; the downstream application applies source priority to select which definition the end user sees.

6 Downstream Application

To validate the dictionary’s practical utility, we built a language pack generator that produces language-specific SQLite databases for two Chinese learning applications. The generator swaps in definitions from the target language, rebuilds search indexes, and generates cross-reference terms. Output sizes range from 61–67 MB (30,546 single-character entries) to 327–384 MB (all 428,073 entries) per language, with build times of 5–140 seconds. The dictionary is currently deployed in production for English and Spanish.

Limitations

Glosses, not definitions. The resource produces translation equivalents (glosses), not full lexicographic definitions with usage notes, example sentences, grammatical patterns, or register markers. Entries lack sense discrimination: a polysemous word like 打 produces a flat list without indicating which target-language verb corresponds to which Chinese sense. For inflecting target languages, bare citation forms lack grammatical metadata (German noun gender, Russian aspect pairs).

Evaluation limitations. The 87.3% sense coverage figure is an automated lexical overlap proxy, not a human judgment, and the evaluation is partially circular (§5). For the six languages without community references, we have no external quality measure. Comparison against MT baselines (NLLB, Google Translate) remains future work.

Pivot translation bias. All translations are mediated through community-language context (primar-

ily English), inheriting English sense boundaries. The LLM may partially bypass this via its multilingual training, but direct Chinese→target evaluation is needed.

Single-model homogeneity. All 5.1M LLM-generated glosses originate from a single model on a single API. Model-specific biases propagate uniformly, and reproducibility depends on this API remaining available.

Headword and language selection. The 428K headwords are inherited from CC-CEDICT without curation; the inventory includes phone numbers, anime titles, and proper nouns. The 12 target languages overrepresent European languages (5 of 12) and exclude Hindi, Arabic, Thai, and Malay.

No interchange format. The resource is stored as flat text in SQLite without compliance with TEI-Lex-0, OntoLex-Lemon, or ISO 24613 LMF, limiting interoperability.

7 Future Work

From glossary to dictionary. Sense-discriminated entries mapping each target-language gloss to a specific Chinese sense with usage guidance. LLMs could produce numbered sense-aligned translations, or multilingual embeddings could perform post-hoc alignment.

Pivot translation bias as a window into LLM representations. A controlled experiment comparing pivot-translated glosses against direct translations across typologically diverse languages would test whether LLMs reproduce English sense boundaries or restructure senses from their multilingual training.

Computational loanword detection across the Sinosphere. The dictionary now contains both Hokkien source forms and Southeast Asian target languages—infrastructure for tracing lexical diffusion at scale. Linking Hokkien *tāu-hū* to Indonesian *tahu*, Tagalog *taho*, and Thai ตะฮู้ would produce a machine-readable atlas of Chinese vocabulary diffusion across maritime Southeast Asia (Chan-Yap, 1980).

Under-resourced and endangered languages. The ~\$100 cost makes the pipeline relevant for languages without institutional support for manual lexicography. Within China alone, Zhuang (16M speakers), Tibetan (6M), Uyghur (10M), and Yi (7.8M) lack comprehensive bilingual dictionaries.

Community-in-the-loop evaluation. Native-speaker evaluation across all 12 target languages—

particularly by diaspora communities who are primary users of bilingual dictionaries—would establish ground-truth quality metrics.

8 Conclusion

We have demonstrated that a single commercial LLM can produce a multilingual Chinese glossary of unprecedented scale—428,073 headwords in 12 languages—for approximately \$100 in API inference costs. The key enablers are (1) batched prompting that generates all languages simultaneously, (2) language-specific prompt rules that prevent script contamination, and (3) a context-aware backfill pipeline that ensures complete coverage. The resulting resource, combined with community dictionaries and 184K dialect forms spanning Cantonese and Hokkien, provides the first open multilingual Chinese lexical resource covering major languages of Europe, East Asia, Southeast Asia, and the Middle East.

The resource is a beginning, not an endpoint. Transforming gloss lists into sense-structured, grammatically enriched, human-validated dictionary entries—and extending the approach to under-resourced languages—is a multiyear research program requiring expertise in computational lexicography, multilingual NLP, lexical typology, and community-based language documentation. The scientific questions this work opens—How do LLMs represent cross-lingual lexical semantics? Does pivot translation reproduce or restructure sense boundaries? Can computational methods trace historical loanword diffusion across language families?—are as valuable as the artifact itself. We release the complete resource, build tools, and evaluation data under CC BY-SA 4.0, and invite the community to build upon it.

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A Prompt Templates

A.1 System Prompt (v2, production)

```
You are a professional multilingual
Chinese
lexicographer producing dictionary-
style
definitions in 12 languages.

Rules:
- Output EXACTLY one line per language
  in
```

```

format "xx: def1/def2"
- Be concise: dictionary style, not
  full
  sentences
- Maximum 5 glosses per entry
- CRITICAL: Every non-Chinese
  definition must
  contain ZERO Chinese characters.

Language-specific rules:
- ja: Provide the MEANING in Japanese,
  not
  just kanji echo or kana reading.
- ko: Write in Hangul only. Do NOT mix
  in
  Chinese characters.
- vi: Write in Vietnamese with proper
  diacritics. Do NOT include Chinese
  characters.

```

```

tl: tokwa      <- Hokkien tau-koan
vi: dau phu   <- Sino-Vietnamese
ko: 두부       <- Sino-Korean
ja: とうふ     <- Sino-Japanese

```

This entry illustrates how the Hokkien maritime trade vocabulary seeded cognates across Southeast Asian languages: *tāu-hū* (Hokkien) → *tahu* (Indonesian/Malay), *taho/tokwa* (Tagalog), *đậu phụ* (Vietnamese, Sino-Vietnamese route).

A.2 Backfill Prompt (context-aware)

```

Fill in ONLY the missing language
definitions.
Existing translations are provided for
consistency -- match their style and
specificity.
Do NOT repeat or modify existing
translations.

1. 三角褲 / 三角褲
   Pinyin: san jiao ku
   Existing translations:
     en: briefs/underwear
     de: Slip/Unterhose
     fr: slip/calecon
     [... 8 more languages ...]
   MISSING -- fill these:
   vi:

```

B Sample Definitions

B.1 Idiom: 画蛇添足

huà shé tiān zú “draw legs on a snake”

```

en: to improve something unnecessarily
;
   to gild the lily
de: der Schlange Fuesse hinzumalen;
   etwas Ueberfluessiges tun
fr: dessiner un serpent et lui ajouter
   des pattes; faire du zeile
es: dibujar una serpiente y anadirle
   patas; hacer algo innecesario
ko: 뱀에게 발을 그리다;
   불필요한 것을 추가하다
vi: ve ran them chan;
   them thu khong can thiet

```

B.2 Loanword chain: 豆腐

dòufu “tofu”

```

en: tofu / bean curd
de: Tofu / Bohnenkäse
id: tahu      <- Hokkien tau-hu

```